UbugeniPalace

2025

UBUGENI PALACE

Discover Authentic Rwandan Craftsmanship

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UbugeniPalace

Artisan Marketplace

Web Design Final Exam Project Documentation

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1 Project Introduction

1.1 Title

UbugeniPalace - Digital Marketplace for Rwandan Artisans

1.2 Case Study

UbugeniPalace is a web-based e-commerce platform designed to empower Rwandan artisans by providing a digital marketplace to showcase and sell their authentic handmade crafts to customers worldwide. The platform addresses challenges faced by local artisans, such as limited market access and high commission fees on global ecommerce platforms, by creating a dedicated space that celebrates Rwandan craftsmanship and culture.

1.3 Purpose

The primary purpose of UbugeniPalace is to bridge the gap between traditional Rwandan artisans and global customers, fostering economic empowerment and cultural preservation. The platform solves the following problems:

- **Market Access:** Enables artisans to reach customers beyond local markets.
- **Cultural Preservation:** Showcases the stories and cultural significance behind each craft.
- **Economic Empowerment:** Provides artisans with a fair platform to retain more profits.
- **User-Friendly Experience:** Offers an intuitive interface for customers to discover and purchase unique products.

1.4 Technologies Used

- **Frontend:** HTML5, CSS3, JavaScript (ES6+)
- **Backend:** PHP 8.0+
- **Database:** MySQL 8.0+
- **Server:** Apache (via XAMPP/WAMP)
- **Additional Technologies:** AJAX for dynamic content loading, JSON for data storage, CSS Grid and Flexbox for responsive design

2 Problem Statement

2.1 The Problem

Rwandan artisans, who create high-quality handmade crafts such as pottery, baskets, and jewelry, face significant barriers in reaching broader markets:

1. **Limited Market Reach:** Artisans rely on local markets, craft fairs, or word-ofmouth, restricting their customer base.
2. **High Platform Fees:** Existing e-commerce platforms charge substantial commissions, reducing artisans profits.
3. **Lack of Digital Presence:** Many artisans lack the technical skills to create and manage online stores.
4. **Cultural Disconnect:** Generic platforms fail to highlight the cultural significance and stories behind Rwandan crafts.
5. **Customer Accessibility:** Customers seeking authentic Rwandan products struggle to find reliable sources online.

2.2 Target Users

- **Artisans:** Traditional craft makers, modern artists, and small craft businesses seeking a digital platform to sell their products.
- **Customers:** Local and international customers, including Rwandans in the diaspora, tourists, and gift buyers who value authentic, handmade products.
- **Admins:** Platform administrators responsible for managing users, approving products, and monitoring system performance.

2.3 Solution Features

UbugeniPalace addresses these challenges with the following features:

- **Digital Marketplace:** A user-friendly platform for artisans to list and sell products.
- **Cultural Storytelling:** Artisan profiles that highlight personal stories and craft heritage.
- **Fair Commission Structure:** Lower fees compared to major e-commerce platforms.
- **Bilingual Support:** Content in English and Kinyarwanda to cater to local and international users.
- **Secure Transactions:** Safe payment processing and order tracking for customers.
- **Advanced Search and Filtering:** Easy product discovery with category, price, and artisan-based filters.

3 System Requirements

3.1 Software Requirements

- **Web Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari (latest versions).
- **Code Editor:** Visual Studio Code or any modern code editor.
- **Local Server Environment:** XAMPP or WAMP with Apache and MySQL.
- **PHP Version:** 8.0 or higher.
- **MySQL Version:** 8.0 or higher.

3.2 Hardware Requirements (Optional)

- **Processor:** Intel i3 or equivalent.
- **RAM:** Minimum 4GB, recommended 8GB.
- **Storage:** 10GB free space for project files and server environment.
- **Internet Connection:** Required for development, testing, and accessing external resources.

3.3 Development Environment Setup

1. **Install XAMPP/WAMP:** Download and install from https://www.apachefriends. org/.
2. **Start Services:** Launch Apache and MySQL services via the XAMPP/WAMP control panel.
3. **Database Setup:** Create a database named ubumenyi_bwubugeni and import the schema from database/ubumenyi_database.sql.
4. **Project Setup:** Place the project folder in the htdocs directory of XAMPP/WAMP.
5. **Configure Database:** Update credentials in config/database.php if necessary.

4 System Design

4.1 User Flow Diagram

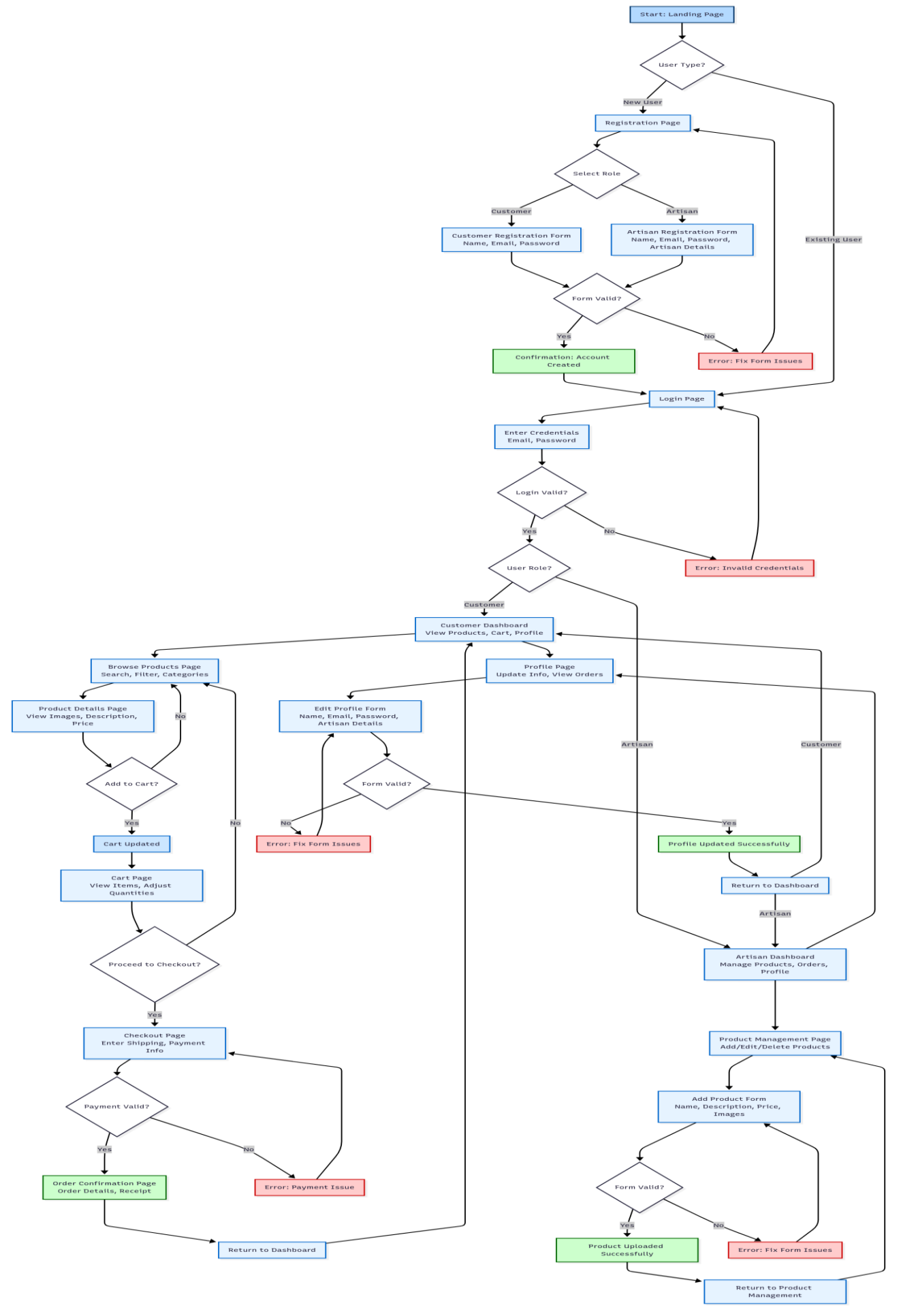
Customer Flow:

Homepage Browse Products Filter/Search Product Details Add to Cart/Wishlist Check

Artisan Flow:

Register Create Profile Upload Products Manage Inventory Track Orders View Analyt

Admin Flow: Login Dashboard Manage Users/Products Approve Listings Monitor Orders View Analytic


Start: Landing Page
User Type?
New User
Existing User
Registration Page
Select Role
Customer
Artisan
Customer Registration Form
Name, Email, Password
Artisan Registration Form
Name, Email, Password,
Artisan Details
Form Valid?
Yes
No
Confirmation: Account
Created
Error: Fix Form Issues
Login Page
Enter Credentials
Email, Password
Login Valid?
Yes
No
Error: Invalid Credentials
User Role?
Customer
Artisan
Customer Dashboard
View Products, Cart, Profile
Browse Products Page
Search, Filter, Categories
Profile Page
Update Info, View Orders
Product Details Page
View Images, Description,
Price
No
Edit Profile Form
Name, Email, Password,
Artisan Details
Add to Cart?
Yes
No
Form Valid?
No
Yes
Error: Fix Form Issues
Cart Updated
Profile Updated Successfully
Cart Page
View Items, Adjust
Quantities
Return to Dashboard
Artisan
Proceed to Checkout?
Artisan Dashboard
Manage Products, Orders,
Profile
Yes
Checkout Page
Enter Shipping, Payment
Info
Product Management Page
Add/Edit/Delete Products
Payment Valid?
Add Product Form
Name, Description, Price,
Images
Yes
No
Order Confirmation Page
Order Details, Receipt
Error: Payment Issue
Form Valid?
Yes
No
Product Uploaded
Successfully
Error: Fix Form Issues
Return to Dashboard
Return to Product
Management

4.2 System Architecture

Frontend (HTML5, CSS3, JavaScript)
(HTTP
Requests/AJAX) Backend
(PHP)
(PDO Queries)
Database (MySQL)

4.3 Page Structure

The website consists of the following main pages:

1. **Homepage** (index.php): Features hero section, featured products, and artisan highlights.
2. **Products** (pages/products.php): Displays product catalog with search and filtering options.
3. **Artisan Directory** (pages/artisans.php): Lists artisans with profile details and their products.
4. **Product Details** (pages/product-details.php): Shows detailed product information, images, and reviews.
5. **User Authentication** (pages/login.php, pages/register.php): Handles user registration and login.
6. **User Dashboard** (pages/dashboard.php): Personalized area for managing orders, wishlist, and profile.
7. **Shopping Cart & Checkout** (pages/cart.php, pages/checkout.php): Manages cart and order placement.
8. **Admin Panel** (admin/): Provides tools for user management, product approval, and analytics.

4.4 Navigation Structure

- **Main Navigation:** Home, Products, Artisans, About, Contact
- **User Menu:** Login/Register, Dashboard, Profile, Logout
- **Admin Menu:** Dashboard, Users, Products, Orders, Analytics
- **Mobile Navigation:** Responsive, touch-friendly menu for mobile devices

5 Implementation

5.1 HTML Implementation

- **Semantic Structure:** Used HTML5 elements (<header>, <main>, <section>, <article>, <footer>) for accessibility and SEO.

- **Form Design:** Created user-friendly forms for registration, login, product uploads, and checkout with proper labeling and input validation.
- **Accessibility Features:** Included alt text for images, ARIA attributes, and proper heading hierarchy.
- **SEO Optimization:** Implemented meta tags, structured data, and clean URLs for better search engine visibility.

5.2 CSS Implementation

- **Modern CSS:** Utilized CSS Variables, Flexbox, and Grid for consistent and responsive layouts.
- **Mobile-First Design:** Ensured compatibility across devices with media queries and responsive units (rem, vw, %).
- **Animations:** Added smooth transitions, hover effects, and loading animations for enhanced user experience.
- **File Organization:**
  - assets/css/style.css: Core styles and theming.
  - assets/css/responsive.css: Device-specific styles.
  - assets/css/animations.css: Animation and transition effects.

5.3 JavaScript Implementation

- **Interactivity:** Implemented dynamic features like search suggestions, image galleries, and mobile menu toggles using ES6+.
- **AJAX:** Used Fetch API for asynchronous content loading (e.g., search suggestions, cart updates).
- **Form Validation:** Added client-side validation for forms (e.g., email format, required fields) with server-side backup.
- **File Organization:**
  - assets/js/main.js: Core functionality and initialization.
  - assets/js/cart.js: Shopping cart interactions.
  - assets/js/validation.js: Form validation logic.
  - assets/js/animations.js: Animation scripts.

5.4 PHP Implementation

- **User Authentication:** Built secure login/logout system with Bcrypt password hashing and session management.
- **Database Operations:** Used PDO with prepared statements for secure CRUD operations.
- **File Uploads:** Implemented secure image upload system with validation for type, size, and storage.
- **API Endpoints:** Created RESTful APIs for dynamic features like search, cart management, and product uploads.
- **Security Measures:** Included input sanitization, SQL injection prevention, and XSS protection.

5.5 Key Features Implemented

1. **User System:** Multi-role registration/login for customers, artisans, and admins.
2. **Product Management:** Artisans can upload products with multiple images, descriptions, and pricing.
3. **Shopping Cart:** Add/remove items, update quantities, and persist cart across sessions.
4. **Search and Filtering:** Advanced search with category, price, and artisan filters.
5. **Image Galleries:** Display multiple product images with zoom functionality.
6. **Admin Panel:** Tools for managing users, approving products, and tracking orders.
7. **Wishlist System:** Allows customers to save products for later purchase.
8. **Bilingual Support:** English and Kinyarwanda content for accessibility.

6 Database Design

6.1 Entity Relationship Diagram

Users (1) (1) Artisans
Users (1) (M) Orders
Artisans (1) (M) Products
Categories (1) (M) Products
Orders (1) (M) Order_Items

Products (1) (M) Reviews
Users (1) (M) Cart Users (1)
(M) Wishlist

6.2 Database Tables

1. **users**

- id (INT, Primary Key)
- username (VARCHAR(50), Unique)
- email (VARCHAR(100), Unique)
- password (VARCHAR(255), Hashed)
- full_name (VARCHAR(100))
- phone (VARCHAR(20))
- user_type (ENUM: ‘customer’, ‘artisan’, ‘admin’)
- profile_image (VARCHAR(255))
- address (TEXT)
- created_at (TIMESTAMP)
- updated_at (TIMESTAMP)
- is_active (BOOLEAN) 2. **artisans** id (INT, Primary Key) user_id (INT, Foreign Key users.id) bio (TEXT)
- specialization (VARCHAR(100))
- experience_years (INT)
- location (VARCHAR(100))
- workshop_address (TEXT)
- social_media (JSON)
- rating (DECIMAL(3,2))
- total_reviews (INT)

is_featured (BOOLEAN) cover_image (VARCHAR(255))

3. **categories**

- id (INT, Primary Key)
- name (VARCHAR(50))

- name_kinyarwanda (VARCHAR(50))
- description (TEXT)
- image (VARCHAR(255))
- is_active (BOOLEAN)
- sort_order (INT)

4. **products**

- id (INT, Primary Key)
- artisan_id (INT, Foreign Key artisans.id)
- category_id (INT, Foreign Key categories.id)
- name (VARCHAR(100))
- name_kinyarwanda (VARCHAR(100))
- description (TEXT)
- price (DECIMAL(10,2))
- discount_price (DECIMAL(10,2)) stock_quantity (INT) materials (VARCHAR(255)) dimensions (VARCHAR(100))
- weight (DECIMAL(5,2))
- colors (VARCHAR(255))
- main_image (VARCHAR(255))
- gallery_images (JSON)
- is_featured (BOOLEAN)
- status (ENUM: ‘active’, ‘inactive’, ‘sold’)
- views_count (INT)

5. **orders**

- id (INT, Primary Key)
- user_id (INT, Foreign Key users.id)
- order_number (VARCHAR(20), Unique)
- total_amount (DECIMAL(10,2))

- shipping_fee (DECIMAL(10,2))
- tax_amount (DECIMAL(10,2))
- status (ENUM: ‘pending’, ‘processing’, ‘shipped’, ‘delivered’, ‘cancelled’)
- payment_status (ENUM: ‘pending’, ‘paid’, ‘failed’)
- shipping_address (JSON)
- created_at (TIMESTAMP)

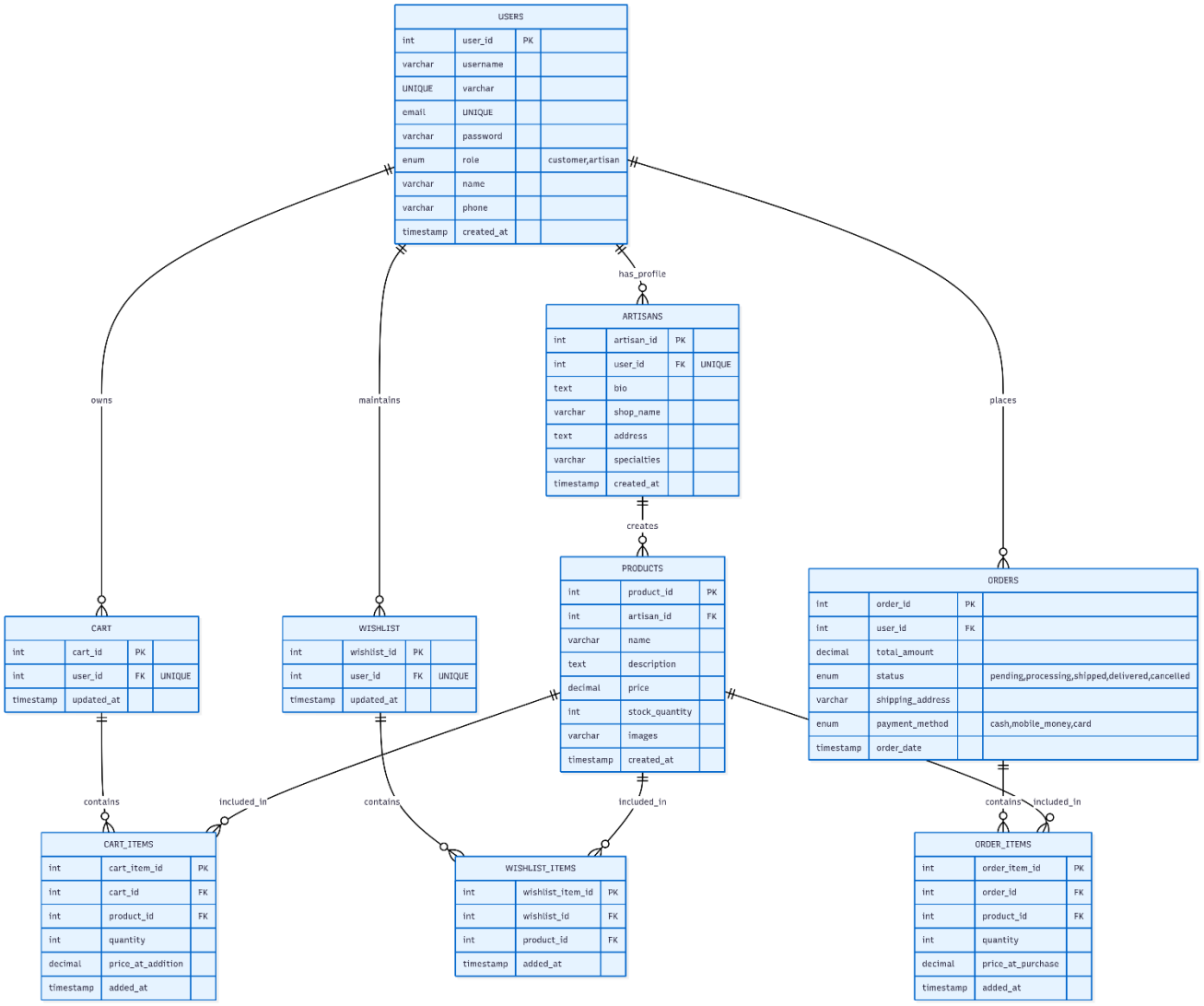
6. **order_items**
   - id (INT, Primary Key)
   - order_id (INT, Foreign Key orders.id)
   - product_id (INT, Foreign Key products.id)
   - quantity (INT)
   - unit_price (DECIMAL(10,2))
   - total_price (DECIMAL(10,2))
7. **cart** id (INT, Primary Key) user_id (INT, Foreign Key users.id) product_id (INT, Foreign Key products.id)
   - quantity (INT)
   - custom_notes (TEXT)
   - added_at (TIMESTAMP)
8. **reviews**
   - id (INT, Primary Key)
   - user_id (INT, Foreign Key users.id) product_id (INT, Foreign Key products.id) rating (INT, 1-5)
   - comment (TEXT)
   - created_at (TIMESTAMP)
9. **wishlist**
   - id (INT, Primary Key)

- user_id (INT, Foreign Key users.id)
- product_id (INT, Foreign Key products.id)
- added_at (TIMESTAMP)

6.3 Database Relationships

- **One-to-One:** Users Artisans (each artisan is linked to a single user account).
- **One-to-Many:** Users Orders, Artisans Products, Categories Products.
- **Many-to-Many:** Users Products (via cart, wishlist, and reviews tables).
- **Data Integrity:** Foreign key constraints, unique constraints on usernames and emails, and cascade deletes ensure data consistency.

6.4 ER diagram For the Database



7 Testing

7.1 Testing Methodology

The project was thoroughly tested to ensure functionality, usability, and security across all user roles and features. Testing was conducted manually, focusing on user workflows, data validation, and system performance.

7.2 User Role Testing

1. **Customer Testing:**

    - **Registration:** Successfully created customer accounts with valid and invalid inputs.
    - **Product Browsing:** Tested search, filtering by category, price, and artisan name.

Shopping Cart/Wishlist: Added/removed items, updated quantities, and verified persistence.

Checkout Process: Completed orders and verified order confirmation.

Reviews: Submitted ratings and comments for purchased products.

2. **Artisan Testing:**
   - **Registration:** Created artisan accounts with profile details and verified admin approval.
   - **Product Upload:** Uploaded products with multiple images and validated stock management.
   - **Order Management:** Processed customer orders and updated statuses.
   - **Analytics:** Viewed sales and product view statistics.
3. **Admin Testing:**
   - **User Management:** Approved/rejected artisan registrations and managed customer accounts.
   - **Product Approval:** Reviewed and approved product listings.
   - **Order Tracking:** Monitored order statuses and processed updates.
   - **Analytics:** Accessed system-wide statistics and reports.

7.3 Cross-Role Testing

- **Customer-Artisan Interaction:** Verified review submissions and direct communication features.

Order Flow: Tested end-to-end order process from customer purchase to artisan fulfillment.

- **Data Consistency:** Ensured data integrity across user actions (e.g., cart updates reflected in orders).

7.4 Bug Discovery and Resolution

1. **Image Upload Issue:**
   - **Problem:** Uploaded images were not displaying due to incorrect file paths.
   - **Solution:** Corrected path configurations and implemented robust file validation.
   - **Impact:** Ensured reliable image display across the platform.
2. **Database Retrieval Issue:**

   Problem: Slow retrieval of product data due to unoptimized queries.

   Solution: Added indexes to frequently queried columns and optimized JOIN operations.

   - **Impact:** Improved page load times and system performance.

3. **UI Positioning Issue:**
    - **Problem:** Dropdown menus were obscured by other elements.
    - **Solution:** Adjusted CSS z-index and positioning properties.
    - **Impact:** Enhanced user interface consistency.
4. **Form Validation Issue:**
    - **Problem:** Client-side validation allowed some invalid submissions.
    - **Solution:** Strengthened JavaScript validation and added server-side checks.
    - **Impact:** Improved data quality and security.

7.5 Device and Browser Testing

- **Browsers:** Tested on Google Chrome and Microsoft Edge for compatibility.
- **Devices:** Verified functionality on desktop, tablet, and mobile devices (320px1920px).
- **Responsiveness:** Ensured layouts and interactions were optimized for all screen sizes.
- **Performance:** Confirmed fast loading times and smooth animations.

7.6 Security Testing

- **Authentication:** Tested login/logout functionality and session expiration.

  Authorization: Verified role-based access restrictions (e.g., admin-only pages).
- **Input Validation:** Attempted submissions with invalid/malicious data to ensure sanitization.
- **File Upload Security:** Tested unauthorized file types and sizes, confirming restrictions.

8 Challenges Faced

8.1 Technical Challenges

1. **Database Design:**

   Challenge: Designing a flexible database to handle diverse craft types while maintaining performance.

   Solution: Used JSON fields for dynamic data (e.g., gallery images) and indexed frequently queried columns.

- **Learning:** Balanced normalization with flexibility in database design.

2. **User Authentication:**
   - **Challenge:** Implementing secure, multi-role authentication with different permissions.
   - **Solution:** Used Bcrypt for password hashing, role-based access control, and admin approval workflows.
   - **Learning:** Gained expertise in secure authentication and authorization.
3. **File Uploads:**
   - **Challenge:** Managing secure uploads of multiple images for product galleries.
   - **Solution:** Implemented validation for file types, sizes, and storage paths.
   - **Learning:** Learned secure file handling and user-generated content management.
4. **Responsive Design:**
   - **Challenge:** Ensuring seamless user experience across various devices.
   - **Solution:** Adopted mobile-first design with CSS Grid, Flexbox, and media queries.
   - **Learning:** Mastered modern CSS techniques and responsive design principles.

8.2 Learning Curve Challenges

1. **Advanced JavaScript:**
   - **Challenge:** Implementing dynamic features like search suggestions and AJAX requests.
   - **Solution:** Studied ES6+ features and used Fetch API for asynchronous operations.
   - **Learning:** Improved proficiency in modern JavaScript and asynchronous programming.
2. **CSS Animations:**
   - **Challenge:** Creating smooth, professional animations without performance issues.
   - **Solution:** Used CSS transitions and keyframes, optimized with Intersection Observer.
   - **Learning:** Learned advanced CSS animation techniques and performance optimization.
3. **Complex Queries:**
   - **Challenge:** Writing efficient MySQL queries for complex filtering and joins.

     Solution: Optimized queries with proper JOINs, indexing, and prepared statements.

     Learning: Enhanced skills in database query optimization and performance tuning.

4. **Project Management:**
    - **Challenge:** Balancing feature development with time constraints.
    - **Solution:** Prioritized core features, used iterative development, and leveraged AI tools for efficiency.
    - **Learning:** Developed skills in project planning and time management.

8.3 Solutions Implemented

- **AI Assistance:** Used AI tools like Cursor AI to accelerate coding and debugging while focusing on learning core concepts.
- **Iterative Development:** Built and tested features incrementally to ensure a functional system at each stage.

Documentation: Maintained clear code comments and organized documentation for easier maintenance and debugging.

9 Conclusion

9.1 Learning Outcomes

- **Technical Skills:**
    - Mastered full-stack development with HTML5, CSS3, JavaScript, PHP, and MySQL.
    - Learned to design scalable database structures with proper relationships and optimization.
    - Implemented secure authentication, data validation, and file handling.
    - Developed responsive, mobile-first interfaces with modern CSS techniques.
- **Problem-Solving Skills:**
    - Identified and addressed a real-world community problem through technology.
    - Developed systematic approaches to debugging and resolving technical issues.
    - Balanced functionality with user experience in design decisions.

Professional Development:

- Created comprehensive documentation and user guides.
- Wrote clean, maintainable, and well-documented code.

- Gained experience in systematic testing and quality assurance.
- Learned to integrate AI tools effectively in development workflows.

9.2 Project Impact

UbugeniPalace demonstrates the power of web technology to address real-world challenges:

- **Economic Empowerment:** Provides artisans with a sustainable platform to sell globally.
- **Cultural Preservation:** Promotes Rwandan craftsmanship through storytelling and visibility.
- **Community Development:** Supports local economies by connecting artisans with customers.

Educational Value: Showcases proficiency in full-stack development, making it an ideal portfolio piece.

9.3 Future Improvements

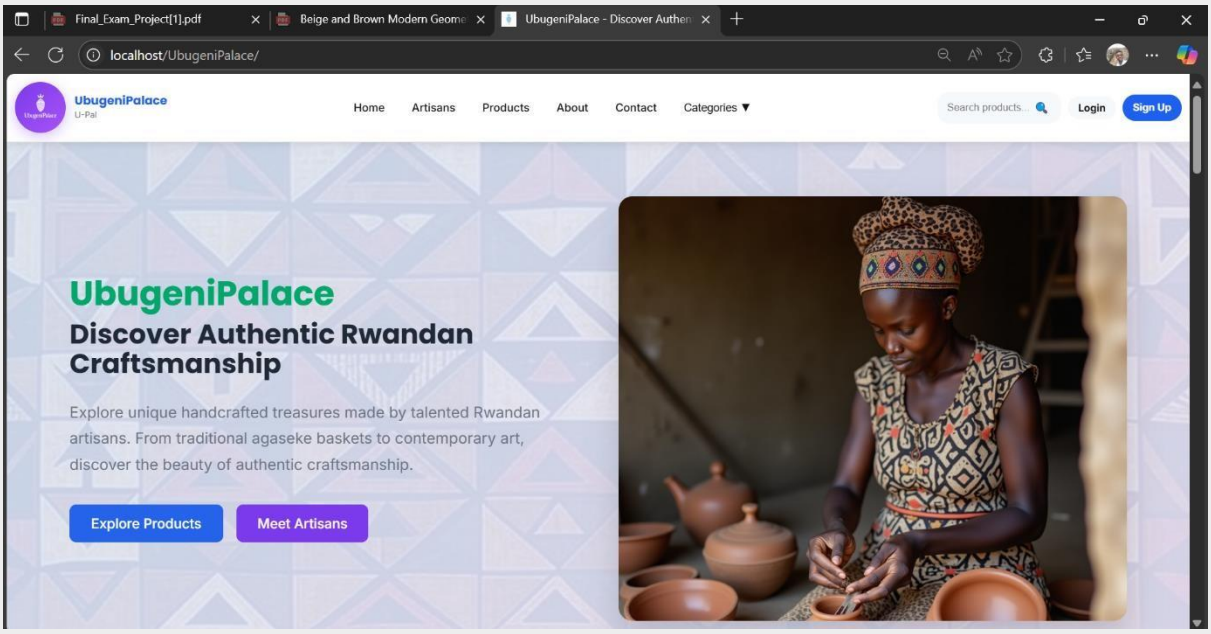
- **Immediate Enhancements:**
  - Integrate payment gateways (e.g., MoMo, Visa, Mastercard).
  - Add email notifications for order updates.
  - Enhance search with full-text indexing and advanced filtering.
  - Develop native iOS and Android apps.
  - Support additional languages (French, Swahili).
- **Long-Term Vision:**
  - Expand to other East African markets.
  - Implement real-time chat for customer-artisan communication.
  - Add advanced analytics for artisans and admins.
  - Introduce AI-driven product recommendations.
  - Migrate to cloud hosting for scalability.

9.4 Personal Growth

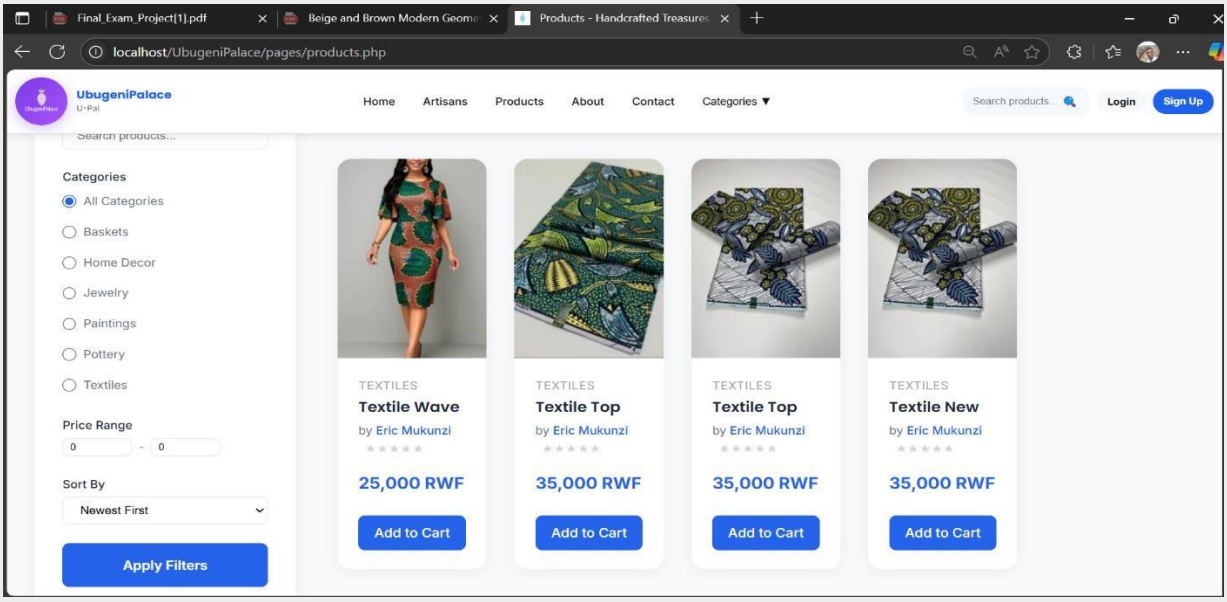
This project has been a transformative experience, enhancing my confidence in full-stack web development. It demonstrated my ability to create a functional, impactful web application that addresses real community needs. The combination of traditional coding skills and modern tools like AI assistance has prepared me for future challenges in software development.

10 Screenshots

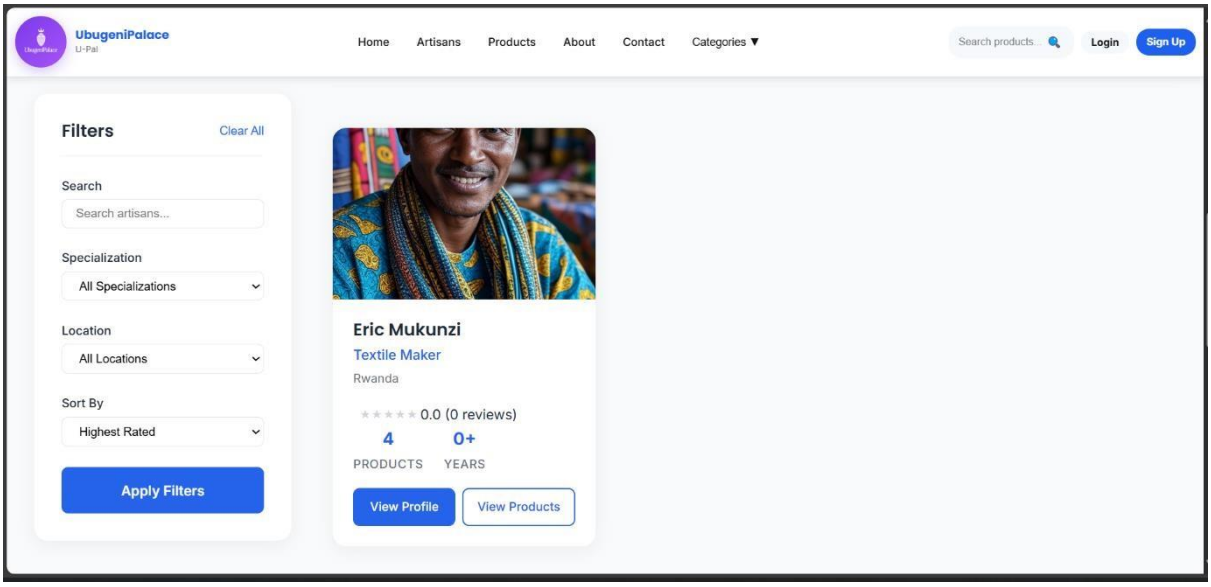
- **Homepage:** Main landing page with hero section and featured products.



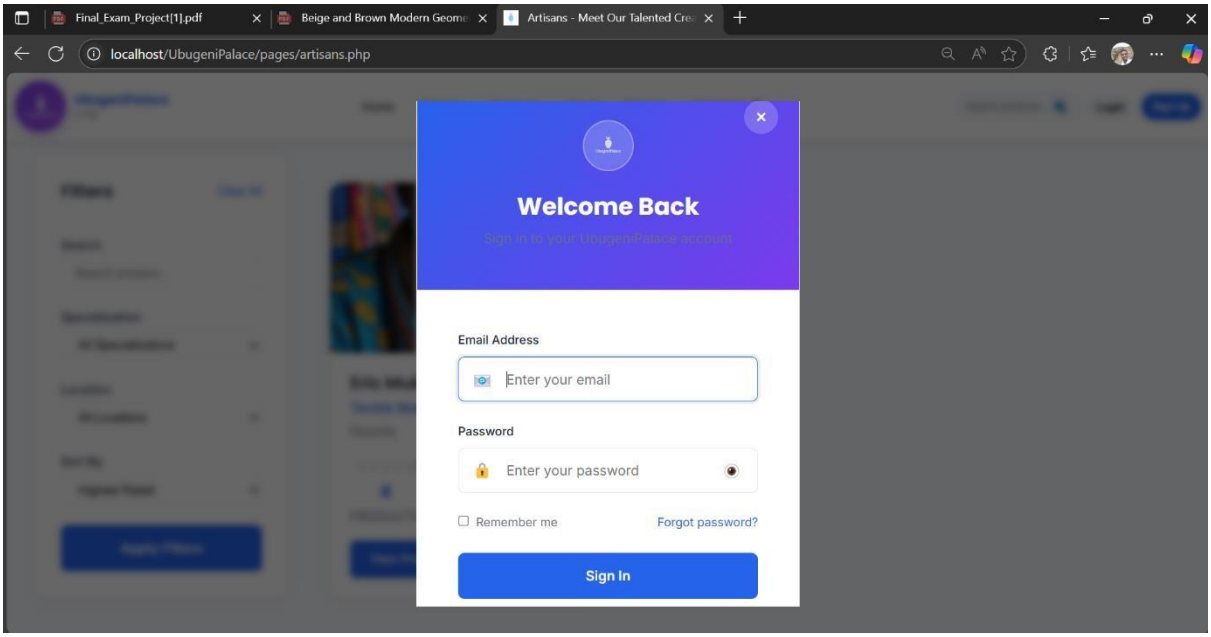
- **Product Catalog:** Product listings with search and filtering options.



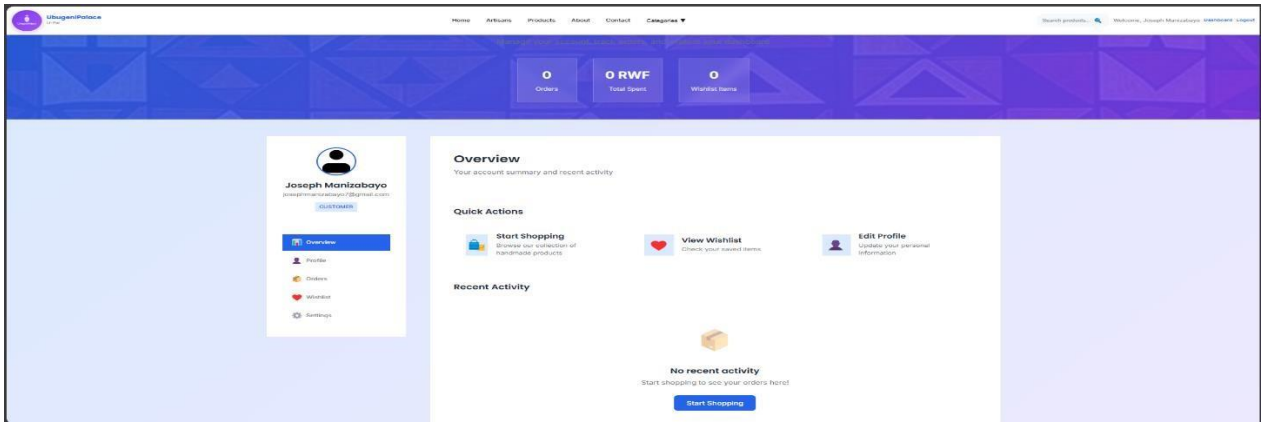
Artisan Directory: Profiles of artisans and their crafts.



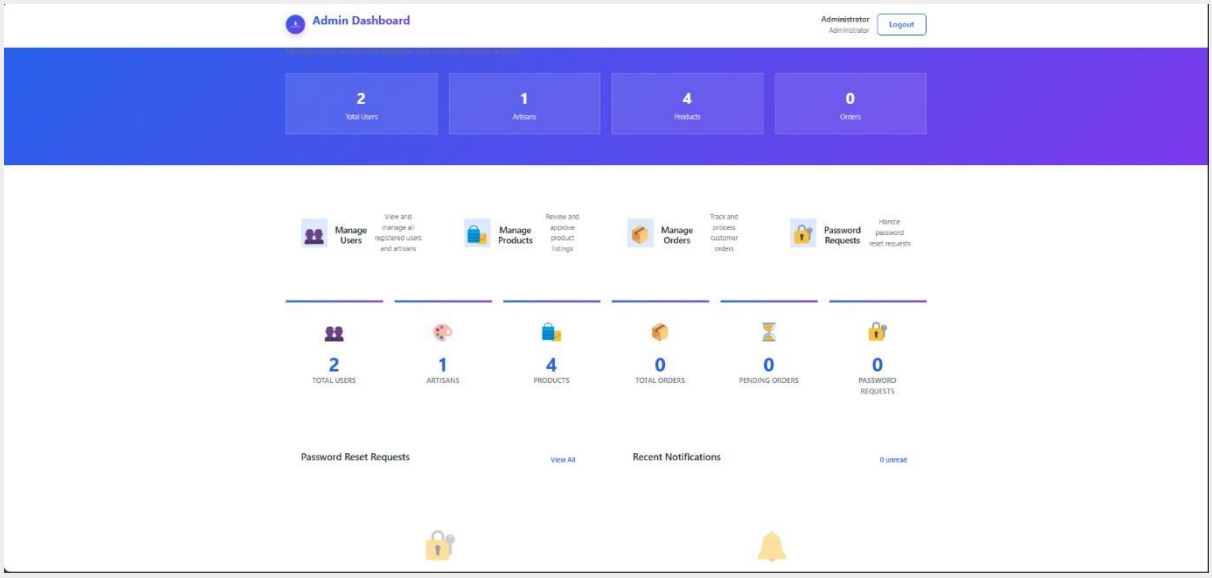
- **User Registration/Login:** Forms for account creation and login.



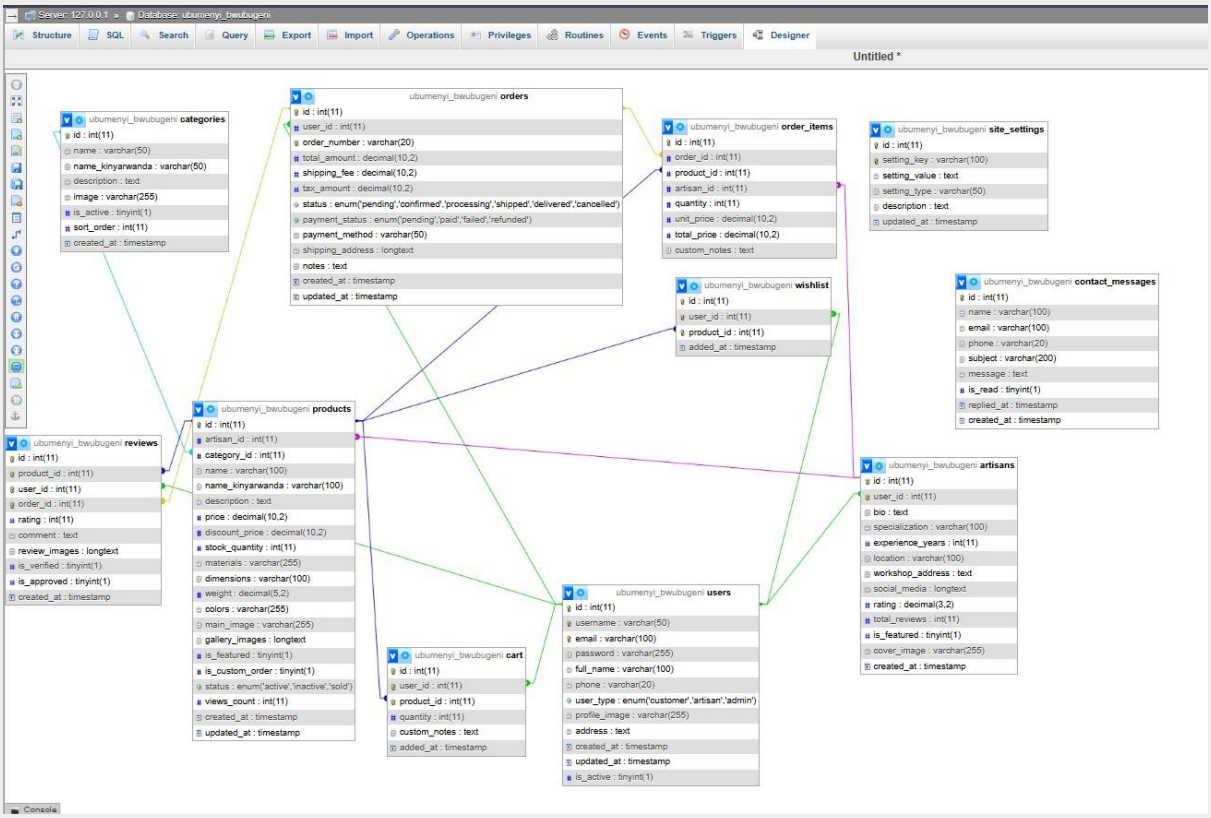
- **User Dashboard:** Personalized area for managing orders and profiles.



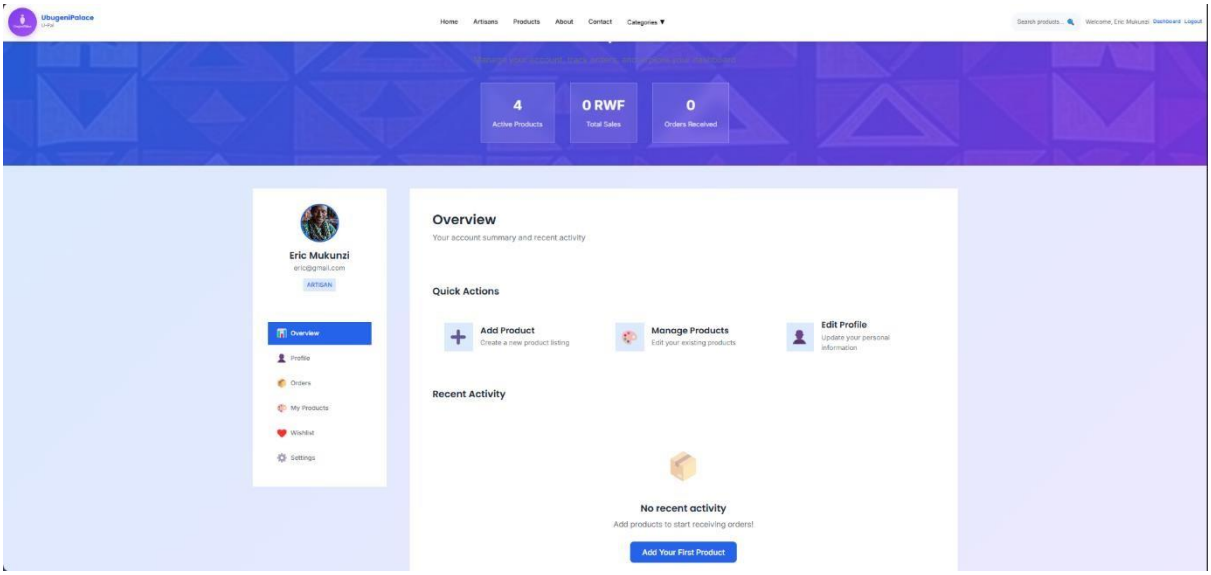
- **Admin Panel:** Dashboard for user and product management.



- **Database Structure:** Visual representation of MySQL tables and relationships.



Artisans Dashboard: Visual representation for an Artisan's dashboard.



Project Summary

- **Project Completion Date:** July 29, 2025
- **Total Development Time:** 4 days
- **Database Tables:** 9 main tables
- **User Roles:** 3 (Customer, Artisan, Admin)
- **Pages Developed:** 15+ main pages
- **Features Implemented:** 20+ core features